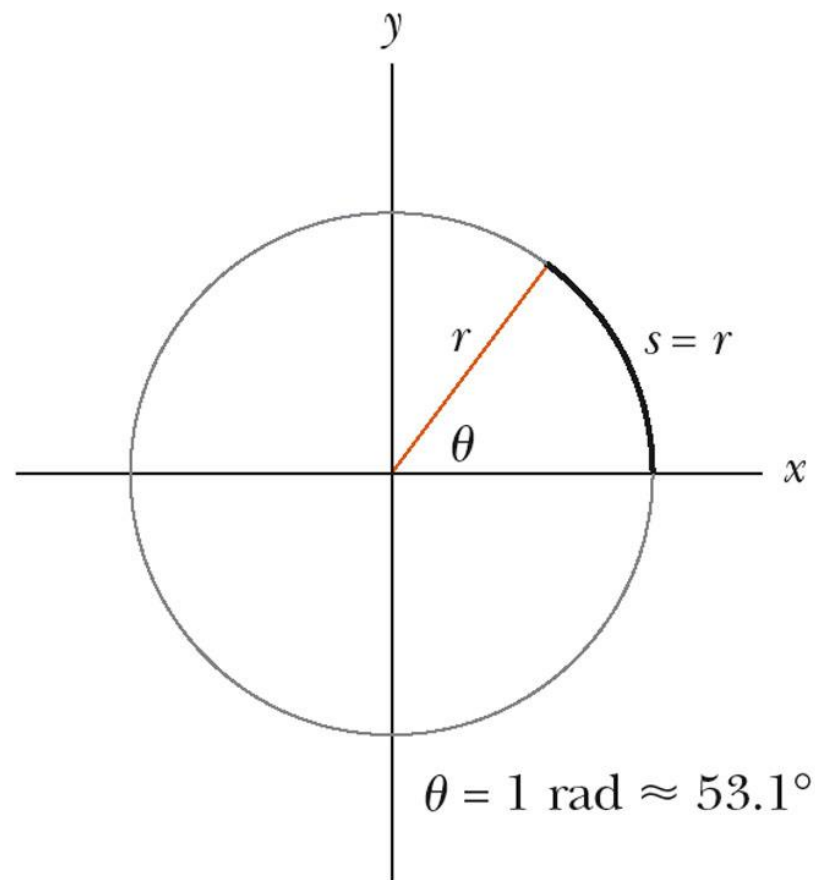


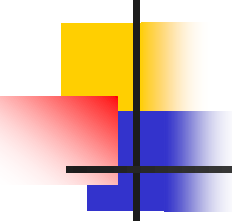
Chapter 7

Rotational Motion and The Law of Gravity

The Radian

- The radian is a unit of angular measure
- The radian can be defined as the arc length s along a circle divided by the radius r
- $\theta = \frac{s}{r}$





More About Radians

- Comparing degrees and radians

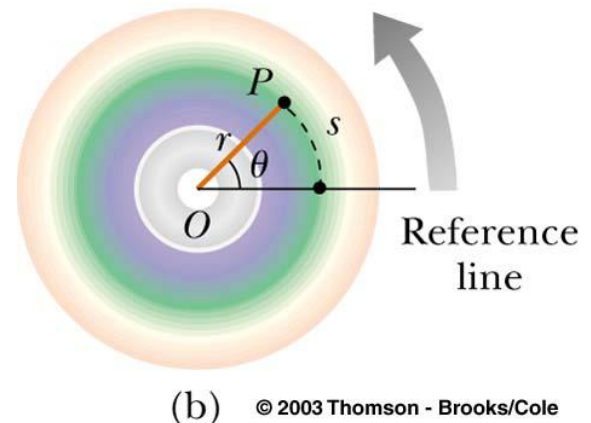
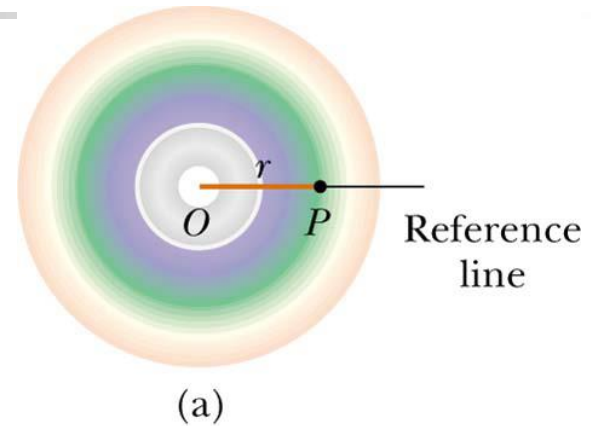
$$1\text{ rad} = \frac{360^\circ}{2\pi} = 57.3^\circ$$

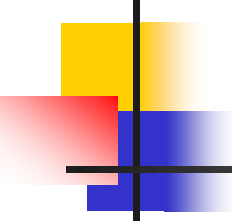
- Converting from degrees to radians

$$\theta [\text{rad}] = \frac{\pi}{180^\circ} \theta [\text{degrees}]$$

Angular Displacement

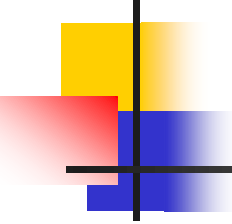
- Axis of rotation is the center of the disk
- Need a fixed reference line
- During time t , the reference line moves through angle θ





Rigid Body

- Every point on the object undergoes circular motion about the point O
- All parts of the object of the body rotate through the same angle during the same time
- The object is considered to be a **rigid body**
 - This means that each part of the body is fixed in position relative to all other parts of the body



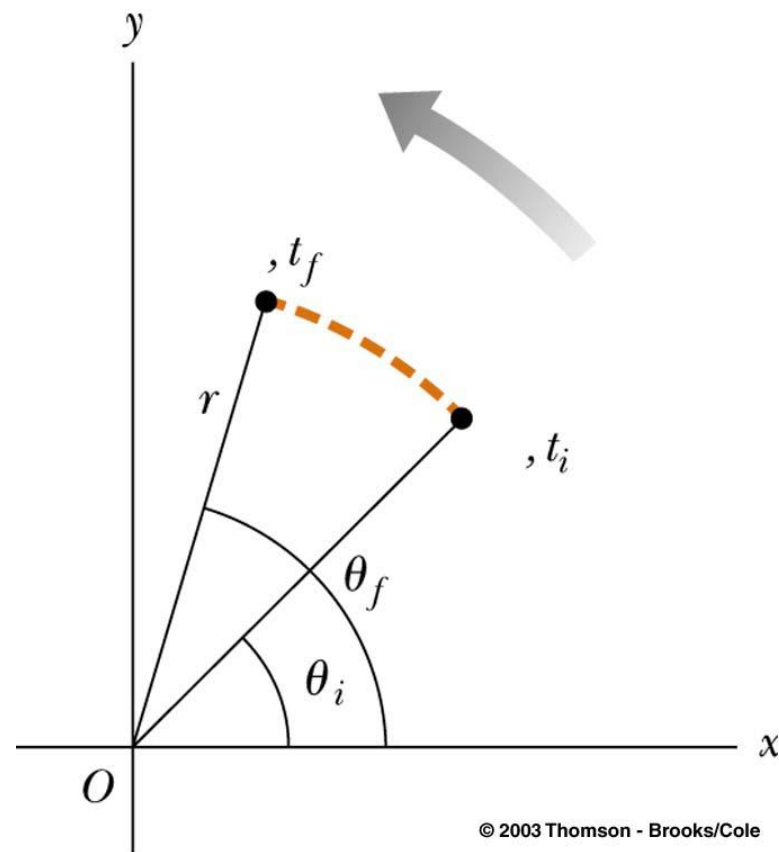
Angular Displacement, cont.

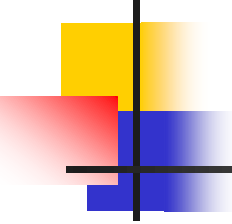
- The *angular displacement* is defined as the angle the object rotates through during some time interval
- $\Delta\theta = \theta_f - \theta_i$
- The unit of angular displacement is the radian
- Each point on the object undergoes the same angular displacement

Average Angular Speed

- The average angular speed, ω , of a rotating rigid object is the ratio of the angular displacement to the time interval

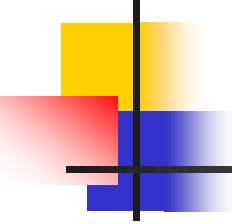
$$\omega_{av} = \frac{\theta_f - \theta_i}{t_f - t_i} = \frac{\Delta\theta}{\Delta t}$$





Angular Speed, cont.

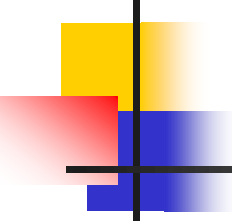
- The *instantaneous* angular speed is defined as the limit of the average speed as the time interval approaches zero
- Units of angular speed are radians/sec
 - rad/s
- Speed will be positive if θ is increasing (counterclockwise)
- Speed will be negative if θ is decreasing (clockwise)



Average Angular Acceleration

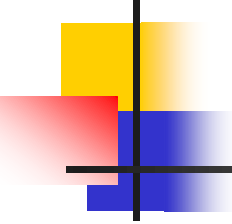
- The average angular acceleration α of an object is defined as the ratio of the change in the angular speed to the time it takes for the object to undergo the change:

$$\alpha_{av} = \frac{\omega_f - \omega_i}{t_f - t_i} = \frac{\Delta\omega}{\Delta t}$$



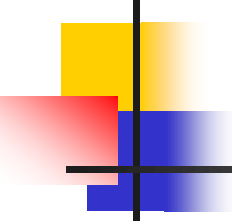
Angular Acceleration, cont

- Units of angular acceleration are rad/s^2
- Positive angular accelerations are in the counterclockwise direction and negative accelerations are in the clockwise direction
- When a rigid object rotates about a fixed axis, every portion of the object has the same angular speed and the same angular acceleration



Angular Acceleration, final

- The sign of the acceleration does not have to be the same as the sign of the angular speed
- The instantaneous angular acceleration is defined as the limit of the average acceleration as the time interval approaches zero



Analogies Between Linear and Rotational Motion

Linear Motion with a Constant
(Variables: x and v)

$$v = v_i + at$$

$$\Delta x = v_i t + \frac{1}{2} at^2$$

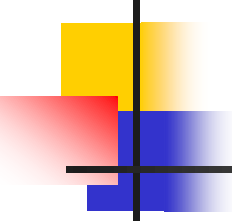
$$v^2 = v_i^2 + 2a\Delta x$$

Rotational Motion about a Fixed Axis with α Constant (Variables: θ and ω)

$$\omega = \omega_i + \alpha t \quad [7.7]$$

$$\Delta \theta = \omega_i t + \frac{1}{2} \alpha t^2 \quad [7.8]$$

$$\omega^2 = \omega_i^2 + 2\alpha\Delta\theta \quad [7.9]$$



Relationship Between Angular and Linear Quantities

- Displacements

$$s = \theta r$$

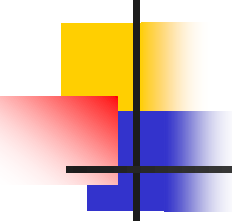
- Speeds

$$v_t = \omega r$$

- Accelerations

$$a_t = \alpha r$$

- Every point on the rotating object has the same angular motion
- Every point on the rotating object does *not* have the same linear motion

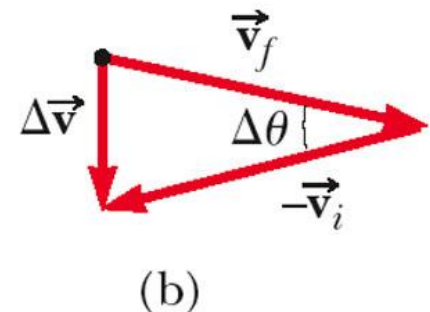
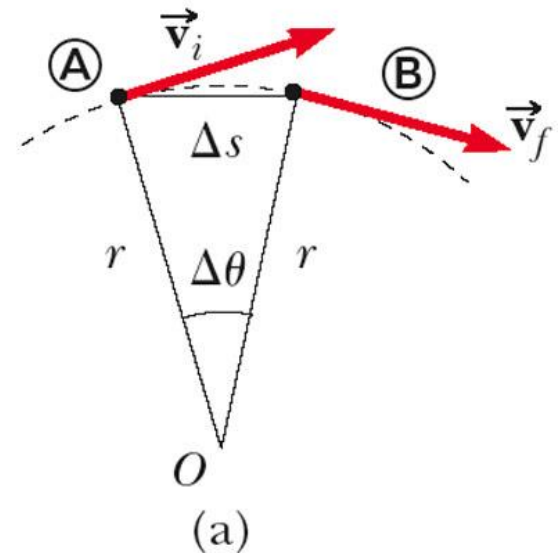


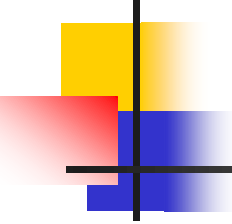
Centripetal Acceleration

- An object traveling in a circle, even though it moves with a constant speed, will have an acceleration
- The centripetal acceleration is due to the change in the *direction* of the velocity

Centripetal Acceleration, cont.

- Centripetal refers to “center-seeking”
- The direction of the velocity changes
- The acceleration is directed toward the center of the circle of motion



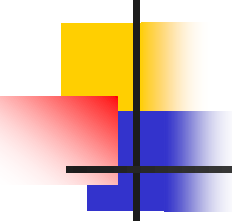


Centripetal Acceleration, final

- The magnitude of the centripetal acceleration is given by

$$a_c = \frac{v^2}{r}$$

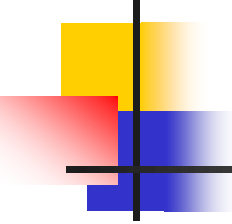
- This direction is toward the center of the circle



Centripetal Acceleration and Angular Velocity

- The angular velocity and the linear velocity are related ($v = \omega r$)
- The centripetal acceleration can also be related to the angular velocity

$$a_c = \omega^2 r$$



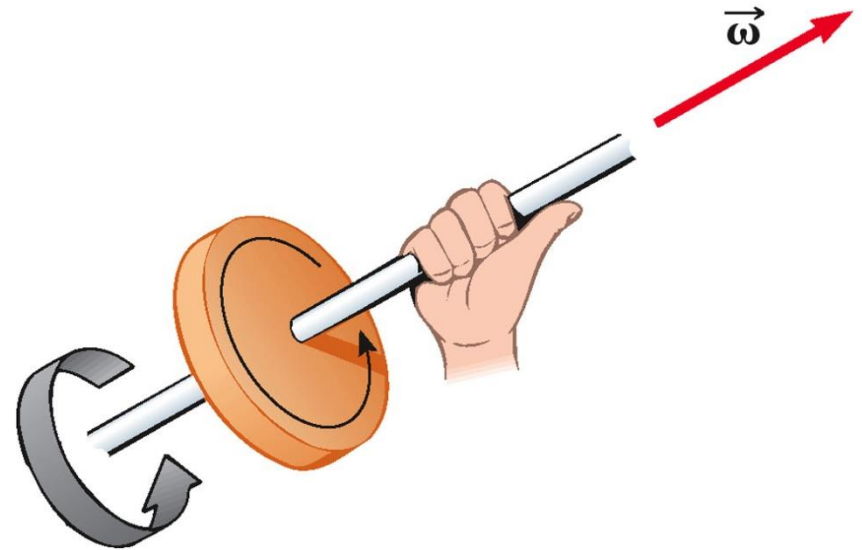
Total Acceleration

- The tangential component of the acceleration is due to changing speed
- The centripetal component of the acceleration is due to changing direction
- Total acceleration can be found from these components

$$a = \sqrt{a_t^2 + a_c^2}$$

Vector Nature of Angular Quantities

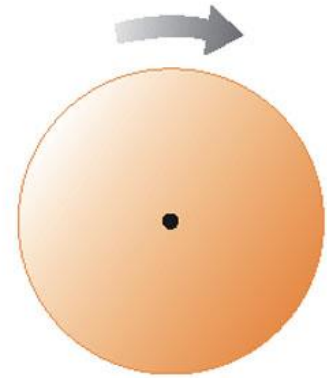
- Angular displacement, velocity and acceleration are all vector quantities
- Direction can be more completely defined by using the right hand rule
 - Grasp the axis of rotation with your right hand
 - Wrap your fingers in the direction of rotation
 - Your thumb points in the direction of ω



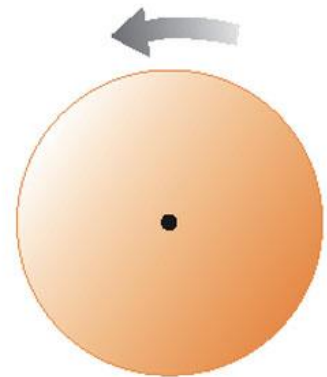
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Velocity Directions, Example

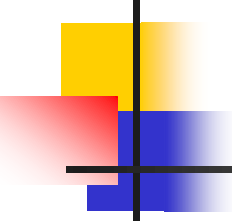
- In a, the disk rotates clockwise, the velocity is into the page
- In b, the disk rotates counterclockwise, the velocity is out of the page



(a)

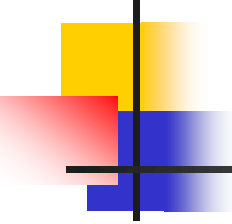


(b)



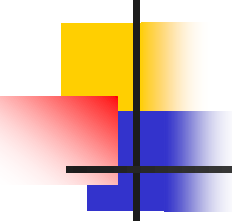
Acceleration Directions

- If the angular acceleration and the angular velocity are in the same direction, the angular speed will increase with time
- If the angular acceleration and the angular velocity are in opposite directions, the angular speed will decrease with time



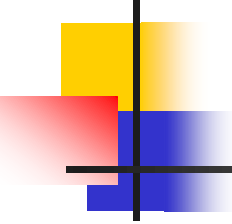
Forces Causing Centripetal Acceleration

- Newton's Second Law says that the centripetal acceleration is accompanied by a force
 - $F_c = ma_c$
 - F_c stands for any force that keeps an object following a circular path
 - Tension in a string
 - Gravity
 - Force of friction



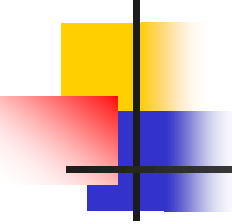
Problem Solving Strategy

- **Draw a free body diagram,** showing and labeling all the forces acting on the object(s)
- **Choose a coordinate system** that has one axis perpendicular to the circular path and the other axis tangent to the circular path
 - The normal to the plane of motion is also often needed



Problem Solving Strategy, cont.

- **Find the net force toward the center** of the circular path (this is the force that causes the centripetal acceleration, F_c)
- **Use Newton's second law**
 - The directions will be radial, normal, and tangential
 - The acceleration in the radial direction will be the centripetal acceleration
- **Solve for the unknown(s)**



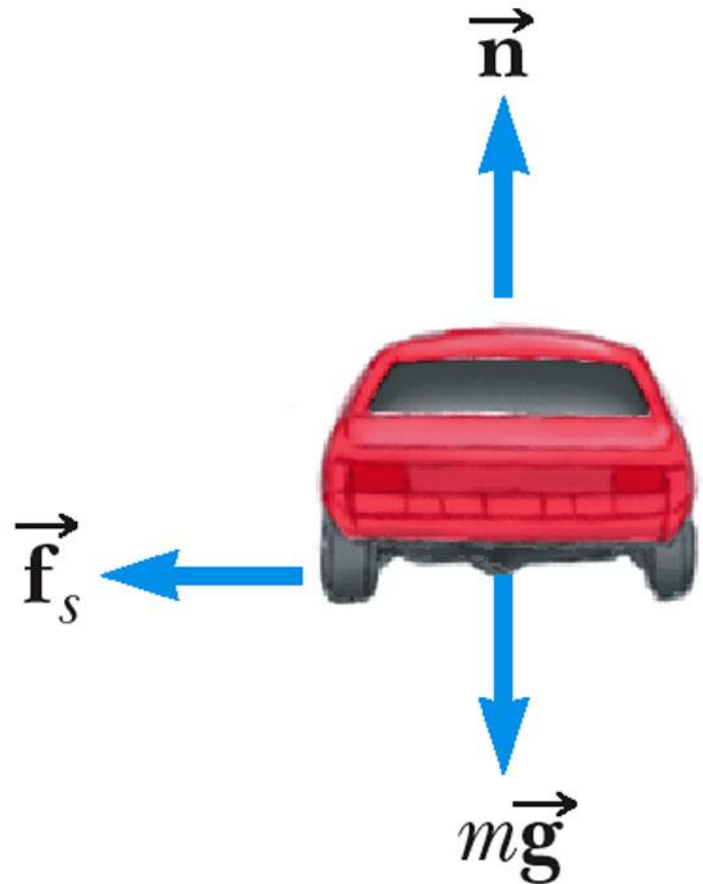
Applications of Forces Causing Centripetal Acceleration

- Many specific situations will use forces that cause centripetal acceleration
 - Level curves
 - Banked curves
 - Horizontal circles
 - Vertical circles

Level Curves

- Friction is the force that produces the centripetal acceleration
- Can find the frictional force, μ , or v

$$v = \sqrt{\mu rg}$$

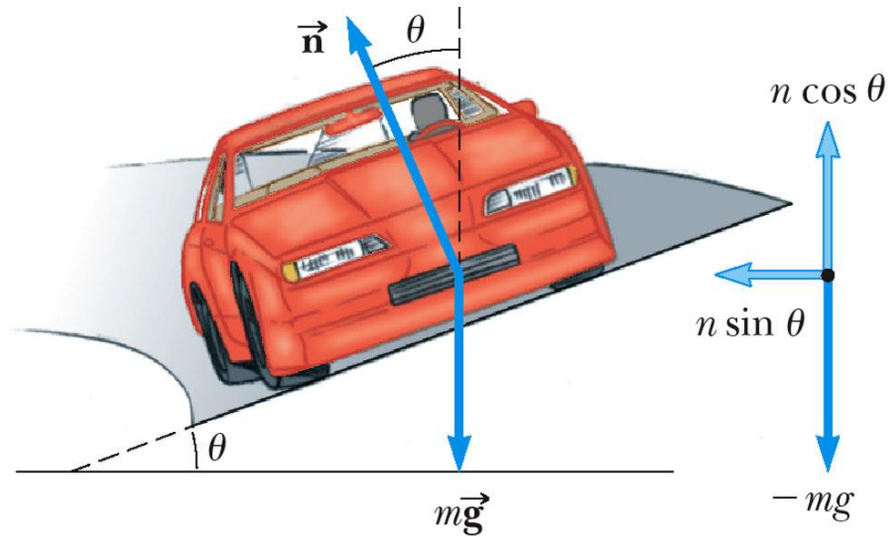


Banked Curves

- A component of the normal force adds to the frictional force to allow higher speeds

$$\tan \theta = \frac{v^2}{rg}$$

$$\text{or } a_c = g \tan \theta$$

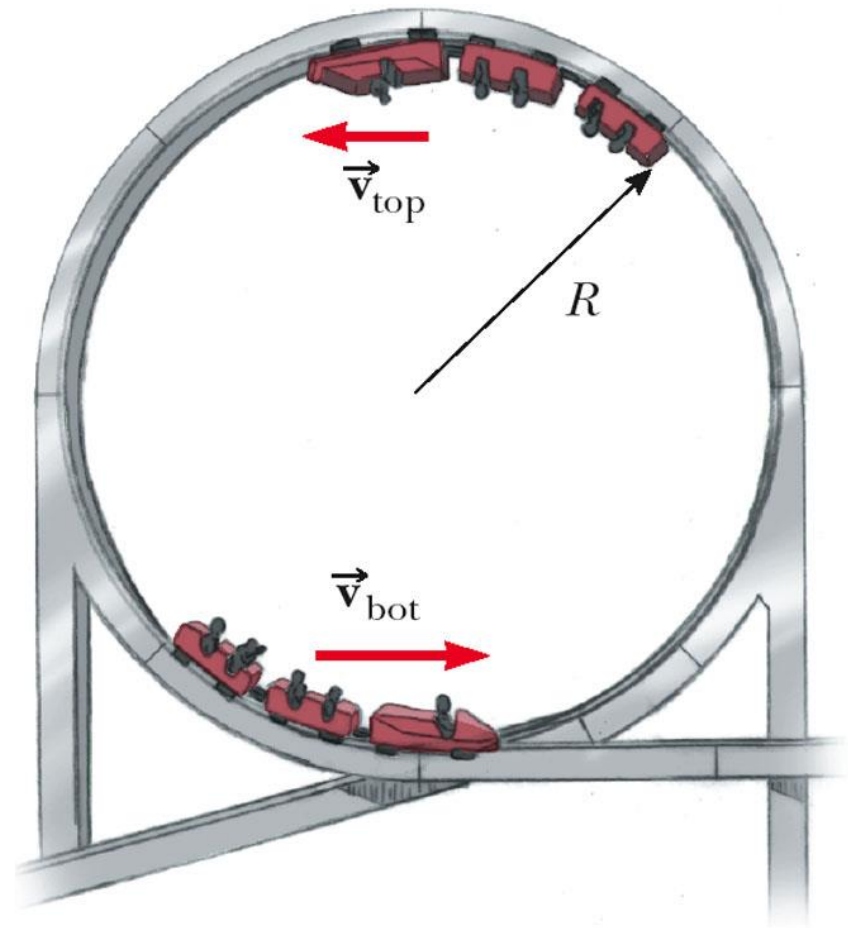


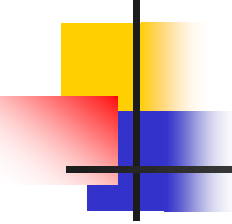
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Vertical Circle

- Look at the forces at the top of the circle
- The minimum speed at the top of the circle can be found

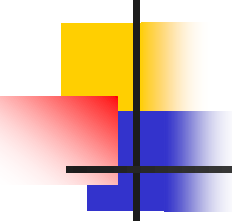
$$v_{\text{top}} = \sqrt{gR}$$





Forces in Accelerating Reference Frames

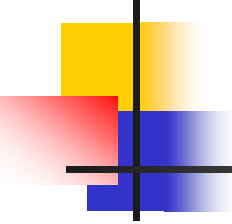
- Distinguish real forces from fictitious forces
- “Centrifugal” force is a fictitious force
- Real forces always represent interactions between objects



Newton's Law of Universal Gravitation

- Every particle in the Universe attracts every other particle with a force that is directly proportional to the product of the masses and inversely proportional to the square of the distance between them.

$$F = G \frac{m_1 m_2}{r^2}$$

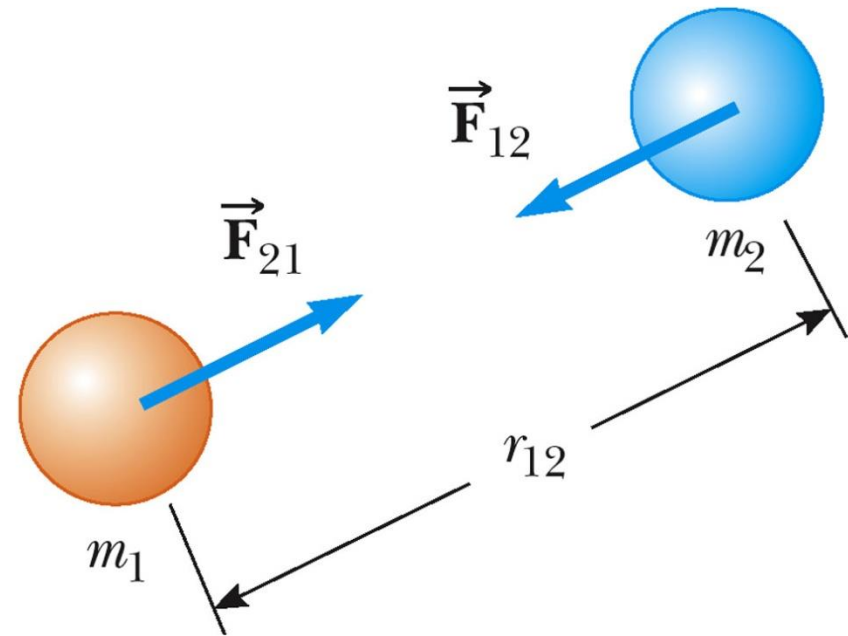


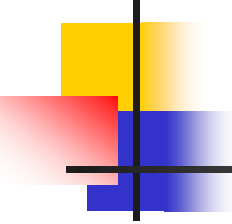
Universal Gravitation, 2

- G is the constant of universal gravitational
- $G = 6.673 \times 10^{-11} \text{ N m}^2 / \text{kg}^2$
- This is an example of an *inverse square law*

Universal Gravitation, 3

- The force that mass 1 exerts on mass 2 is equal and opposite to the force mass 2 exerts on mass 1
- The forces form a Newton's third law action-reaction



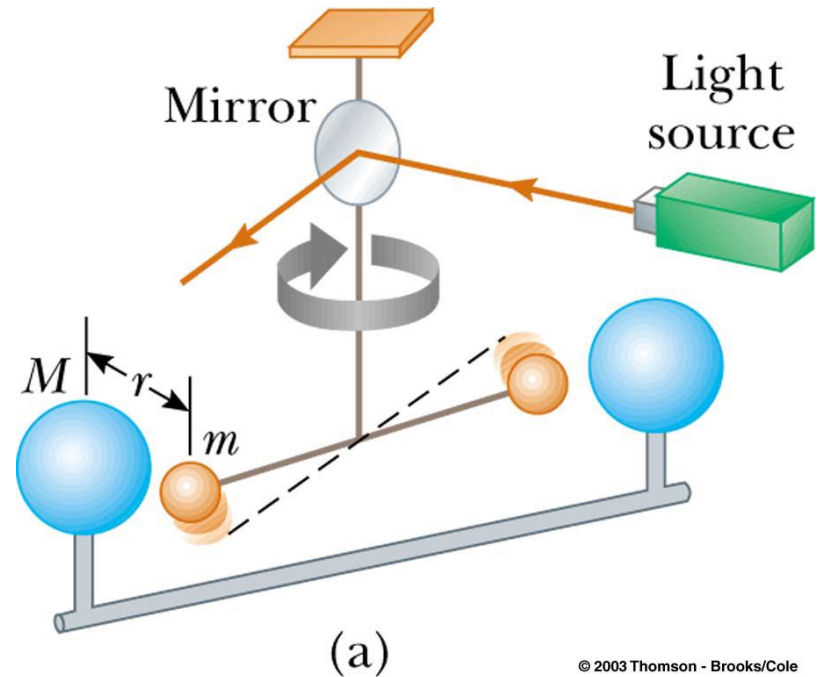


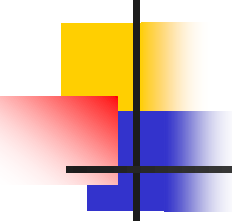
Universal Gravitation, 4

- The gravitational force exerted by a uniform sphere on a particle outside the sphere is the same as the force exerted if the entire mass of the sphere were concentrated on its center
 - This is called Gauss' Law

Gravitation Constant

- Determined experimentally
- Henry Cavendish
 - 1798
- The light beam and mirror serve to amplify the motion





Applications of Universal Gravitation

- Acceleration due to gravity
- g will vary with altitude

$$g = G \frac{M_E}{r^2}$$

TABLE 7.1

Free-Fall Acceleration g at Various Altitudes

Altitude (km) ^a	g (m/s ²)
1 000	7.33
2 000	5.68
3 000	4.53
4 000	3.70
5 000	3.08
6 000	2.60
7 000	2.23
8 000	1.93
9 000	1.69
10 000	1.49
50 000	0.13

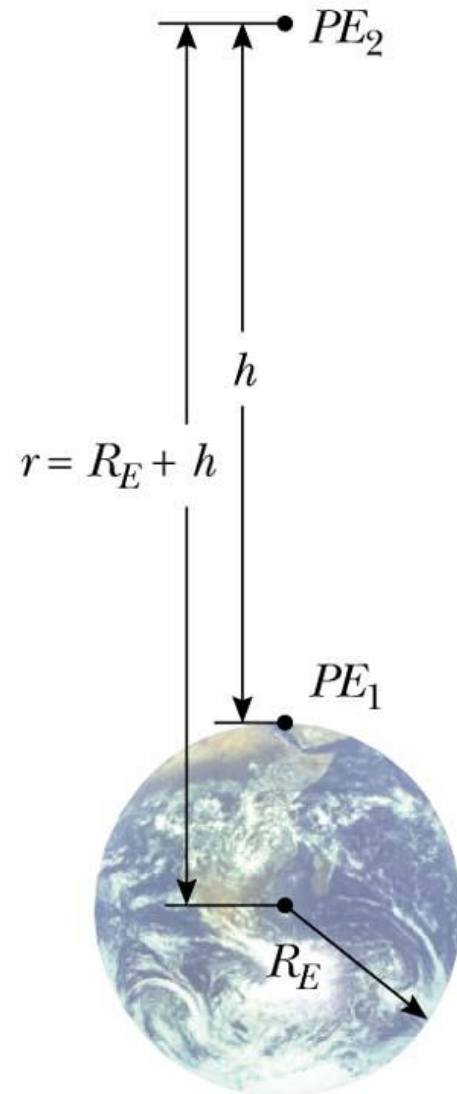
^aAll figures are distances above Earth's surface.

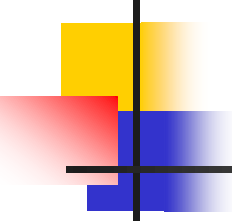
Gravitational Potential Energy

- $PE = mgy$ is valid only near the earth's surface
- For objects high above the earth's surface, an alternate expression is needed

$$PE = -G \frac{M_E m}{r}$$

- Zero reference level is infinitely far from the earth



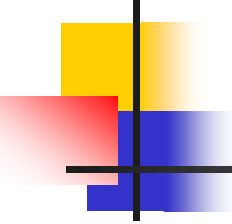


Escape Speed

- The escape speed is the speed needed for an object to soar off into space and not return

$$v_{\text{esc}} = \sqrt{\frac{2GM_E}{R_E}}$$

- For the earth, v_{esc} is about 11.2 km/s
- Note, v is independent of the mass of the object



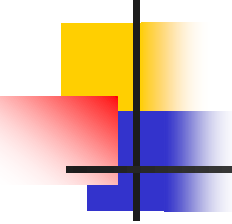
Various Escape Speeds

- The escape speeds for various members of the solar system
- Escape speed is one factor that determines a planet's atmosphere

TABLE 7.2

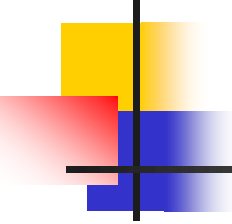
Escape Speeds for the Planets and the Moon

Planet	v_e (km/s)
Mercury	4.3
Venus	10.3
Earth	11.2
Moon	2.3
Mars	5.0
Jupiter	60.0
Saturn	36.0
Uranus	22.0
Neptune	24.0
Pluto	1.1



Kepler's Laws

- All planets move in elliptical orbits with the Sun at one of the focal points.
- A line drawn from the Sun to any planet sweeps out equal areas in equal time intervals.
- The square of the orbital period of any planet is proportional to cube of the average distance from the Sun to the planet.

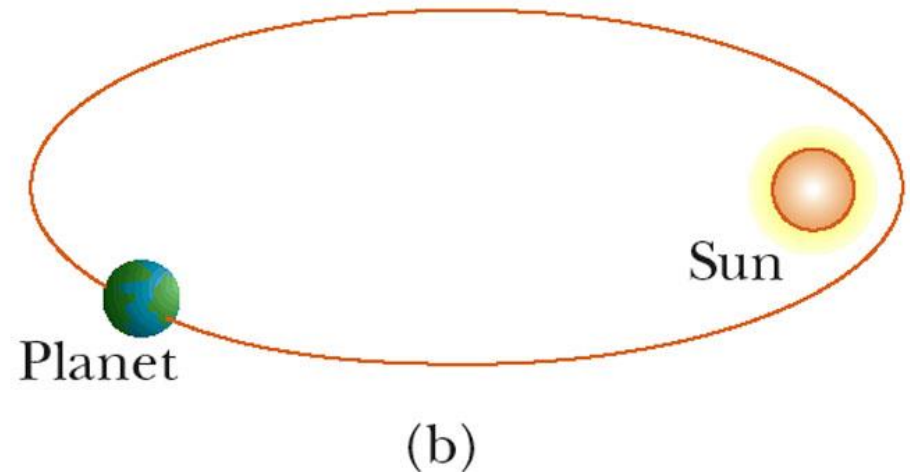
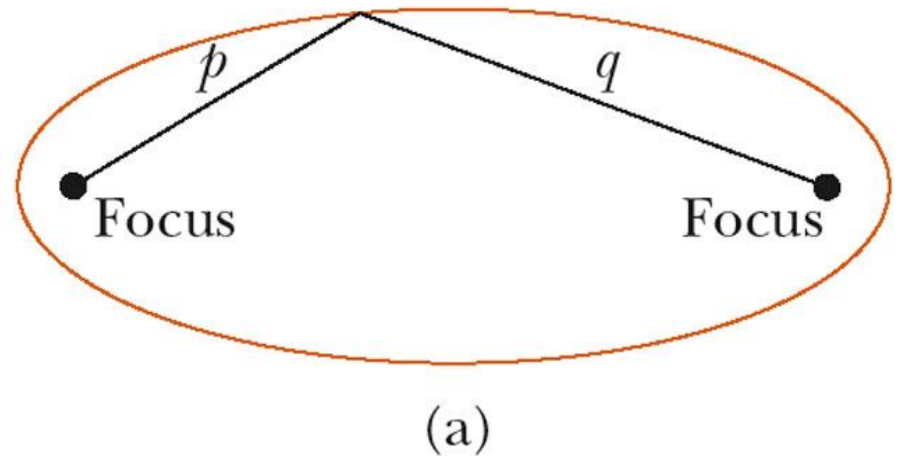


Kepler's Laws, cont.

- Based on observations made by Brahe
- Newton later demonstrated that these laws were consequences of the gravitational force between any two objects together with Newton's laws of motion

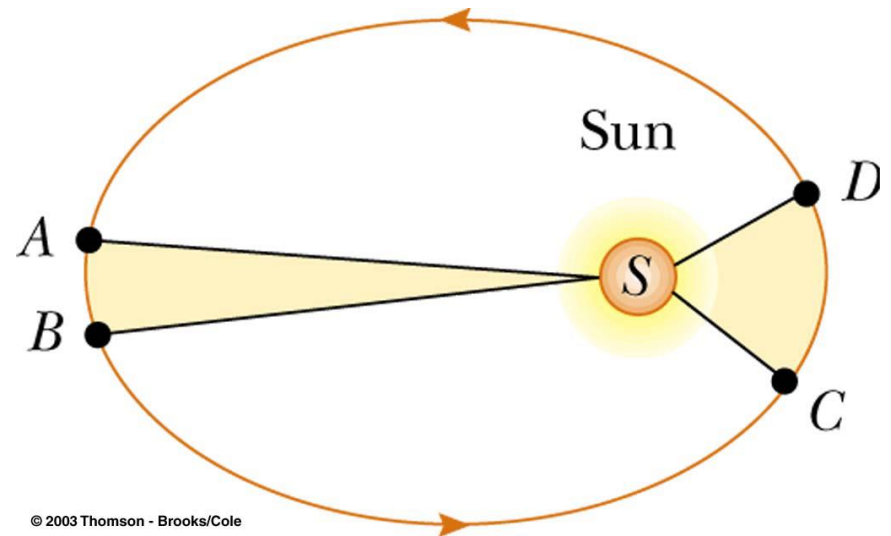
Kepler's First Law

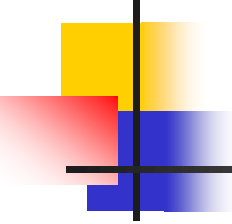
- All planets move in elliptical orbits with the Sun at one focus.
 - Any object bound to another by an inverse square law will move in an elliptical path
 - Second focus is empty



Kepler's Second Law

- A line drawn from the Sun to any planet will sweep out equal areas in equal times
 - Area from A to B and C to D are the same



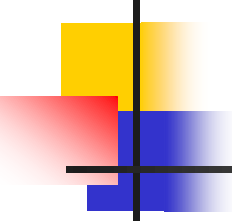


Kepler's Third Law

- The square of the orbital period of any planet is proportional to cube of the average distance from the Sun to the planet.

$$T^2 = Kr^3$$

- For orbit around the Sun, $K = K_S = 2.97 \times 10^{-19} \text{ s}^2/\text{m}^3$
- K is independent of the mass of the planet



Communications Satellite

- A geosynchronous orbit
 - Remains above the same place on the earth
 - The period of the satellite will be 24 hr
- $r = h + R_E$
- Still independent of the mass of the satellite